Delay Prediction in Supply Chains:

A Hybrid Graph-Based and Machine Learning Approach

Candidate: Volpato Mattia

Supervisor: Prof. Michele Ciavotta

Company Co-Supervisor: Eng. Bruno Puzzolante

Master degree

Computer Science

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M4ESTRO Project Overview

Background & Context

Disruptions are reshaping the manufacturing industry, revealing global **supply chain** vulnerabilities and highlighting the critical need for **resilience**.

Supply Chain Challenges

- **Global disruptions** (pandemics, conflicts, climate events) expose supply chain fragility.
- Rising **disruption frequency** and severity create volatile, uncertain, complex, and ambiguous (VUCA) conditions.
- **Resilience** and adaptability are essential for continuity and competitiveness.

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M4ESTRO – A Horizon Europe Project

Project Goals

Develop a resilient, transparent, and adaptive ecosystem under the **Manufacturing-as-a-Service (MaaS)** paradigm, enhancing supply chain flexibility and robustness.

Consortium and Scope

Consortium of **18 partners** from 8 countries, including industrial pilots, technology providers, and research institutions, collaborating over **42 months**.

Strategic Pillars

Four pillars support **M4ESTRO**: resilient manufacturing processes, AI-enabled adaptive equipment, industrial metaverse simulations and human-centered sustainability and resilience.

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Research Motivation & Purpose

M4ESTRO - Industrial Manufacturing As a sErvice STRategies and models for flexible, resilient, and re-configurable value networks through Trusted and Transparent Operations

Motivation

- **Global disruptions** (pandemics, conflicts, climate events) expose supply chain fragility.
- Rising **disruption frequency** and severity create volatile, uncertain, complex, and ambiguous (VUCA) conditions.
- **Resilience** and adaptability are essential for continuity and competitiveness.

Purpose

M4ESTRO introduces the concept of **disruption indicator**:

- **Monitor** the operational **dynamics** of the supply chain.
- **Detect anomalies** in real time across logistics systems.
- **Quantify** the impact of **disruptions** using interpretable data-driven metrics.
- **Enable** timely **responses** through integration with decision-support tools.

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Key Contributions

Supply chain modeling

Mathematical framework for modeling a single-manufacturer **supply chain**.

Indicators Design and Implementation

Logistics Chain Disruption Indicators (LCDIs) to quantify:

- **Dispatch** and **shipment process durations**.
- **Real-time traffic** and **weather** impacts on shipments.
- Effect of **national holidays** on dispatch processes.

Prototype development and deployment

Prototype using pilot partner data **deployed** on AWS cloud infrastructure.

This work addresses Task 1.2, focusing on the design of **Logistics Chain Disruption Indicators (LCDIs)** for detecting ***internal disruptions***.

It concentrates on a subset of LCDIs aimed at **estimating delivery times** and **delays** within a **single-manufacturer supply chain**.

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Dataset & External Sources

Pilot and Tracking Data

Pilot Order Data

FAE Technology, core dataset provider: *one-hundred* order *records* with supplier locations, timestamps and tracking links.

Carrier Tracking Data

17Track API to retrieve *shipment events* across multiple carriers, including *location*, *timestamp*, and *status*.

Geographical Data

Geonames API to standardize shipment events and supplier locations at a *city level*.

Three sources forming the **core dataset**, providing the foundational data that captures the **supply chain structure**.

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Traffic, Weather and Holiday Data

Traffic Data

TomTom API to retrieve *road traffic* data, comparing ideal vs real-time conditions between two locations.

Weather Data

Visual Crossing API to retrieve *weather* data, providing real-time conditions for any location.

Holiday Data

Calendarific API to retrieve *national holidays* worldwide, providing indications on supplier closures.

Three **complementary sources** enrich the core dataset, providing a **complete representation** of the supply chain.

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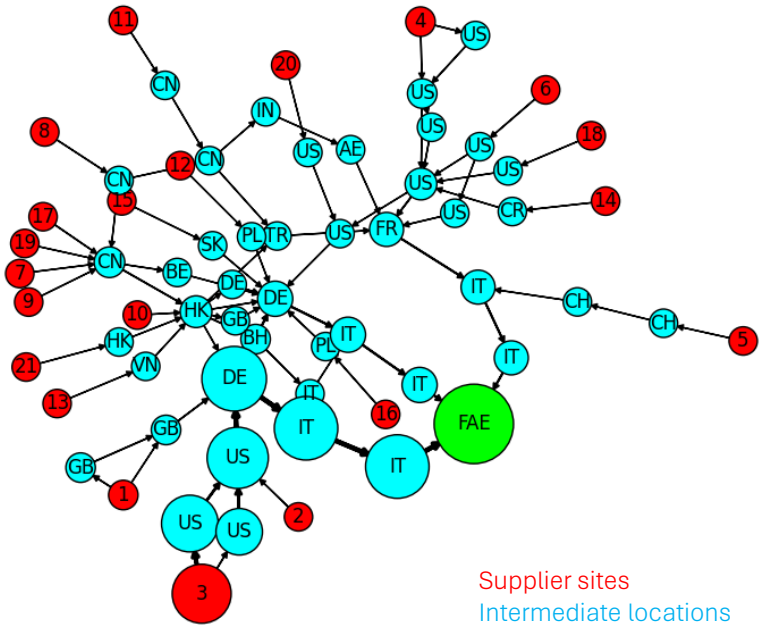
Supply Chain Modeling

Supply Chain Graph Model



Supplier sites
Intermediate locations
Manufacturer

Graph Structure

Directed Acyclic Graph (*DAG*): $G = (V, A, \mathrm{C})$ with C the set of all possible carriers.

Vertices partitioned into:

- Supplier sites (sources).
- Intermediate facilities.
- Manufacturer (sink).

$$V = S \cup I \cup \{m\}$$

Shipments Flow

Flow function indicating the number of shipments transited through a given edge.

$$f : A \to \mathbb{N}$$

Flow conservation on intermediates vertices guarantees no loss or duplication in shipments.

$$\sum_{(i,v)\in A} f(i,v) = \sum_{(v,i)\in A} f(v,i), \qquad \forall i \in I$$

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Markov Chain Routing Model (1)

Empirical Probabilities

Shipment flow function used to estimate the **empirical probabilities** of transition:

$$P_{vu} = \frac{f(v,u)}{f_v(v)} = \frac{f(v,u)}{\sum_{(v,z)\in A} f(v,z)}, \qquad \forall (v,u) \in A$$

The correspondant $P = [P_{vu}]$ forms a row-stocasthic transition probabilities matrix:

$$\sum_{u \in V} P_{vu} = 1\,, \qquad \forall v \in V$$

The manufacturer is treated as an absorbing state:

$$P_{mm} = 1$$

Shipments are represented as **maximal paths** ending in the manufacturer. The **routing process** is modeled using **discrete-time Markov chains**, capturing shipment trajectories as **stochastic processes** based on **empirical transition probabilities** derived from **historical shipment flows**.

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Markov Chain Routing Model (2)

Probabilistic Routing

Given a maximal path:

$$\pi = (v_0, v_1, \dots, v_k = m)$$

The **Markov Property** is used to compute the path probability:

$$\mathbb{P}(\pi) = \mathbb{P}(X_0 = v_0, X_1 = v_1, \dots, X_k = m) = \mathbb{P}(X_0 = v_0) \cdot \prod_{i=0}^{k-1} P_{v_i v_{i+1}}$$

Shipments are represented as **maximal paths** ending in the manufacturer. The **routing process** is modeled using **discrete-time Markov chains**, capturing shipment trajectories as **stochastic processes** based on **empirical transition probabilities** derived from **historical shipment flows**.

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Markov Chain Routing Model (3)

Carrier-Conditioned Dynamics

Shipment flows and routing probabilities are further **conditioned** on the specific **carrier** to model distinct routing behaviors:

$$f^c(v,u) = \text{number of shipments from } v \text{ to } u \text{ executed by carrier } c$$

Vertices without observed carrier-specific transitions default to self-loops to preserve stochasticity:

$$f_v^c(v) = 0 \Leftrightarrow P_{vv}^c = 1$$

Shipments are represented as **maximal paths** ending in the manufacturer. The **routing process** is modeled using **discrete-time Markov chains**, capturing shipment trajectories as **stochastic processes** based on **empirical transition probabilities** derived from **historical shipment flows**.

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Paths Extraction and Probability Distributions

1. Paths Extraction Methodology

Every **maximal path** ends in the manufacturer vertex.

Modified **Depth-First Search** (*DFS*) collects all shipment paths from a vertex v to the manufacturer m.

Despite a possible exponential number of paths, dataset shows only a limited number of routes: **average 4 paths per vertex, maximum 24**.

2. Probability Distribution Computation

Given the collection of all the maximal paths for a vertex $v \in V$:

$$\Pi(v) = \{\pi_1, \pi_2, \dots, \pi_\ell\}$$

The associated paths probabilities conditioned by carrier c:

$$\boldsymbol{p}^c = (p_1^c, p_2^c, \dots, p_\ell^c)$$

Form a valid **probability distribution.**

For each vertex, **carrier behavior** is modeled by considering **all possible paths** to the manufacturer, along with the **probability** that each path is followed.

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Historical Indicators

Historical Indicators Definition

Time Probability Distributions

Dispatch times: a probability distribution for each supplier.

$$\mathrm{DT}\colon S \rightarrow \mathcal{D}(\mathbb{R}_+)$$

Shipment times: a probability distribution for each supplier site–carrier pair.

$$\mathrm{ST}\colon \mathrm{S} \times \mathrm{C} \rightarrow \mathcal{D}(\mathbb{R}_+)$$

Statistical Metrics

Two different metrics for **both operational stages**:

- **Pointwise**: expected value of the distribution.

$$\mathrm{DT}_\mu(\mathrm{s}) = \mathbb{E}[\mathrm{DT}(\mathrm{s})], \qquad \forall \mathrm{s} \in \mathrm{S}$$

- **Interval-based:** β-level confidence interval of the distribution.

$$\mathrm{DT}_\beta(s) = \left[L_\beta(s), U_\beta(s)\right]$$

$$\mathbb{P}\left(L_\beta(s) \leq X \leq U_\beta(s)\right) \geq \beta, \qquad X \sim DT(s)$$

Dispatch process: from order placement to shipment start.
Shipment process: from shipment start to delivery.
Delivery process: combination of dispatch and shipment processes.

Historical indicators are stastistical measures that characterize key **operational times** in the order fulfillment process, based exclusively on **historical data**.

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Historical Indicators Implementation

Theoretical Distribution

Gamma distributions chosen for their established use in supply-chain literature.

Parameter estimation via **Maximum Likelihood Estimation**.

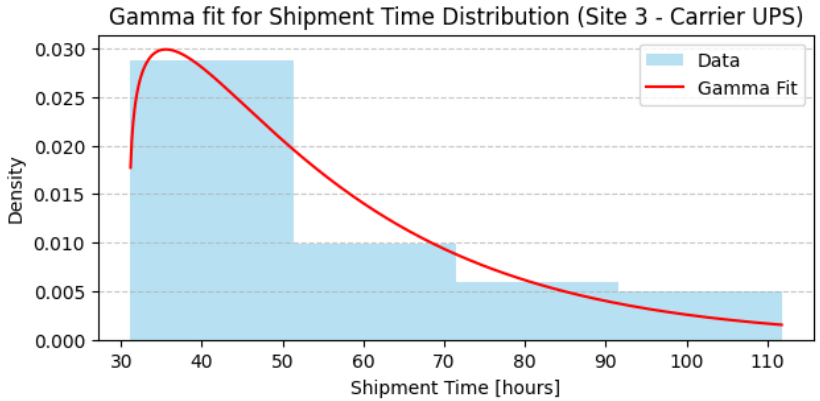
Data threshold: Samples with ≥30 observations.

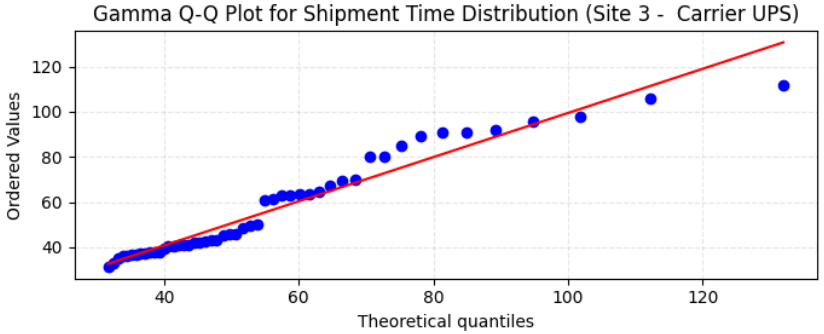
Fitting Procedure

Outlier removal: Interquartile Range (IQR) method.

Goodness-of-fit checks: Q–Q plots and histograms.

Statistical validation: Bootstrap Kolmogorov–Smirnov tests show no evidence against the Gamma assumption.





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Realtime Indicators

Realtime Indicators Overview

Realtime Indicators Hierarchy

Granularity	Scope
Vertex	Supplier and carrier facility.
Route	Carrier route between two locations.
Path	Supplier to manufacturer carrier path.
Graph	Overall supply chain network.

Realtime indicators are statistical measures that provide actionable insights into the current state of the **supply chain.**

Organized **hierarchically** at different **granularity levels**, they combine:

- Historical indicators.
- Supply chain graph.
- Real-time data.
- Realtime indicators from lower levels.

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Realtime Indicators Definition (1)

Vertex Level

Estimated Dispatch Time (EDT_μ): Scalar

- Defined for each **supplier site**.
- Based on the **DT historical indicator**.
- Includes **national holidays** to estimate the dispatch time of a supplier.

$$\mathrm{EDT}_\mu\colon S \to \mathbb{R}_+$$

Vertex Time (VT_β): Interval

- Defined for each **carrier facility**.
- Based on the **historical processing time** distribution of the facility.
- Provides a **β-level confidence interval** of the facility processing time.

$$\mathrm{VT}_\beta\colon I \to \mathbb{R}_+^2, \qquad \mathrm{VT}_\beta = \left(\mathrm{VT}^l, \mathrm{VT}^u\right)$$

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Realtime Indicators Definition (2)

Route Level

Traffic & Weather Meta Index $(\mathrm{TMI}, \mathrm{WMI})$ Scalar

- Based on **real-time data** retrieved from external services.
- Provides a measurement of **traffic** and **weather impacts** on carrier routes.

$$\mathrm{TMI}\colon V^2 \times \mathbb{R}_+ \to [0,1], \qquad \mathrm{WMI}\colon V^2 \times \mathbb{R}_+ \to [0,1]$$

Route Time (RT_β): Interval

- Based on the **historical travel time** distribution of carrier routes.
- Provides a **β-level confidence interval** of the route travel time.

$$\mathrm{RT}_\beta\colon V^2 \to \mathbb{R}_+^2, \qquad \mathrm{RT}_\beta = \left(\mathrm{RT}^l, \mathrm{RT}^u\right)$$

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Realtime Indicators Definition (3)

Route Level

Dynamic Route Time (DRT_β): Interval

- Combines RT_β, TMI and WMI to **include traffic** and **weather information** into the **route time** estimation.

$$\mathrm{DRT}_\beta = f\left(\mathrm{RT}_\beta, \mathrm{TMI}, \mathrm{WMI}\right)$$

$$\mathrm{DRT}_\beta = \left(\mathrm{DRT}^l, \mathrm{DRT}^u\right)$$

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- Realtime indicators from lower levels.

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Realtime Indicators Definition (4)

Path Level

Transit Time (TT_β): Interval

- Based on the **ST historical indicator**.
- Provides a **β-level confidence interval** of the of the **remaining shipment time** without considering the graph structure.

$$\mathrm{TT}_\beta\colon S \times C \times \mathbb{R}_+ \to \mathbb{R}_+^2, \qquad \mathrm{TT}_\beta = (\mathrm{TT}^l, \mathrm{TT}^u)$$

Path Time (PT_β): Interval

- Combines **vertex** and **route time intervals** at **β-level confidence** to estimate the remaining shipment time along a path $\pi = (v_1, v_2, \dots, v_k = m)$.

$$\mathrm{PT}_\beta\colon V^k \times \mathbb{R}_+ \to \mathbb{R}_+^2, \qquad \mathrm{PT}_\beta = (\mathrm{PT}^l, \mathrm{PT}^u)$$

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- Realtime indicators from lower levels.

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Realtime Indicators Definition (5)

Path Level

Total Forecasted Shipment Time (TFST): Interval

- Convex combination of TT_β and PT_β at confidences β_1 and β_2.

$$\mathrm{TFST} = \alpha \cdot \mathrm{TT}_{\beta_1} + (1-\alpha) \cdot \mathrm{PT}_{\beta_2}$$

$$\mathrm{TFST} = (\mathrm{TFST}^l, \mathrm{TFST}^u)$$

Realtime indicators are statistical measures that provide actionable insights into the current state of the **supply chain.**

Organized **hierarchically** at different **granularity levels**, they combine:

- Historical indicators.
- Supply chain graph.
- Real-time data.
- Realtime indicators from lower levels.

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Realtime Indicators Definition (6)

Graph Level

Expected Path Time $(\mathrm{E-PT}_\beta)$: Interval

- Combines the PT_β **estimations** over all the possible **paths** $\boldsymbol{\pi} = (\pi_1, \pi_2, \dots, \pi_\ell)$ from a vertex to the manufacturer according to their likelihood $\mathbf{p}^c = (\mathrm{p}_1, \mathrm{p}_2, \dots, \mathrm{p}_\ell)$.

$$\mathrm{E-PT}_\beta = \sum_{\mathrm{i}=1}^{\ell} p_i^c \cdot \mathrm{PT}_\beta(\pi_i), \qquad \mathrm{E-PT}_\beta = (\mathrm{E-PT}_\beta^l, \mathrm{E-PT}_\beta^{\mathrm{u}})$$

Expected Total Forecasted Shipment Time $(\mathrm{E-TFST})$: Interval

- Extends the **TFST** to consider **all** the **possible paths** from a vertex to the manufacturer.

$$\mathrm{E-TFST} = \alpha \cdot \mathrm{TT}_{\beta_1} + (1-\alpha) \cdot \mathrm{E-PT}_{\beta_2}$$

Realtime indicators are statistical measures that provide actionable insights into the current state of the **supply chain.**

Organized **hierarchically** at different **granularity levels**, they combine:

- Historical indicators.
- Supply chain graph.
- Real-time data.
- Realtime indicators from lower levels.

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Realtime Indicators Implementation (1)

Vertex & Route Time

Ideally VT_β and RT_β would be computed as **confidence intervals** derived from fitted **gamma distributions**.

Due to data scarcity, fitting such distributions was not feasible: we relied on raw **quantiles** of **empirical distributions** instead.

Given a **confidence level** β (or **significance level** $(1-\beta)$), the time interval corresponds to the $\frac{(1-\beta)}{2}$ and $1-\frac{1-\beta}{2}$ **percentiles** of the distribution.

Traffic Meta Index

TMI is implemented only for road transport, calculated as the difference between route time under **actual** and **ideal traffic conditions**.

Weather Meta Index

WMI is implemented through heuristic-based scores that quantify the impact of weather conditions and temperature on carrier transportation.

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TFST Weighting Factor Implementation (1)

Objective

Transit Time (TT): effective for on-time shipments; unable to manage advances or delays.

Path Time (PT): high uncertainty at the start; able to handle advances and delays.

Implement α to:

- **Early phase:** prioritize **Transit Time** to manage higher uncertainty and limited timing deviations.
- **Progressing phase:** prioritize **Path Time** as uncertainty decreases and timing deviations emerge.

Solution

Implement α as a **non-increasing function** of $\tau = \frac{t}{ST_{\mu}(s,c)}$, expressing the **elapsed proportion** of **expected shipment time**.

Three proposed implementations:

1. Constant: $\alpha = k$
2. Linear: $\alpha = max(1 - \tau, 0)$
3. Polynomial: $\alpha(\tau; \langle TT \rangle) = (1 - \tau)^q, \quad q = \frac{1}{\langle TT \rangle} - 1$

$$TFST = \alpha \cdot TT + (1 - \alpha) \cdot PT$$

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TFST Weighting Factor Implementation (2)

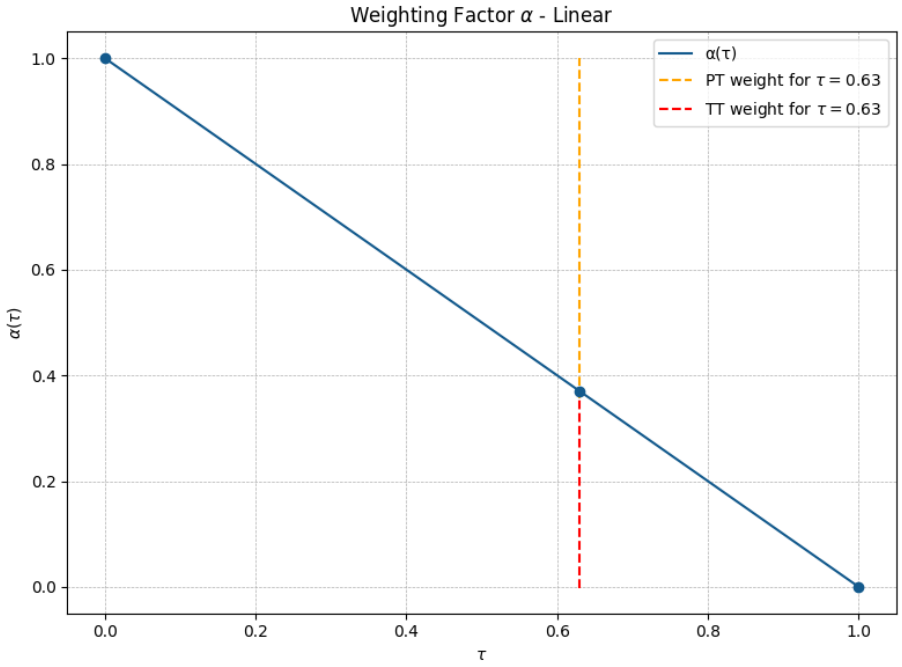
Linear Implementation

Transit Time (TT): Weight decreases linearly as shipment progresses.

Path Time (PT): Weight increases linearly as shipment progresses.

Delayed Shipments: Full weight assigned to PT.

$$\alpha = max(1 - \tau, 0)$$



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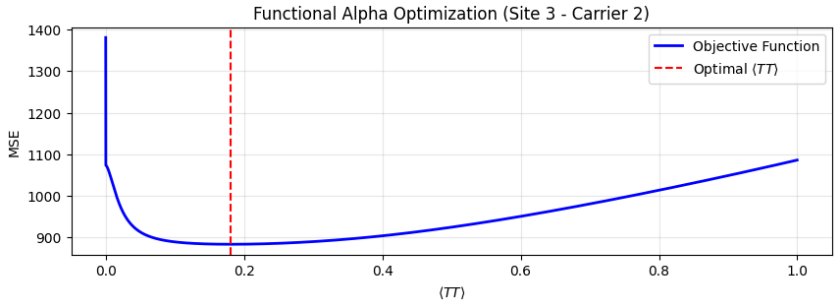
TFST Weighting Factor Implementation (3)

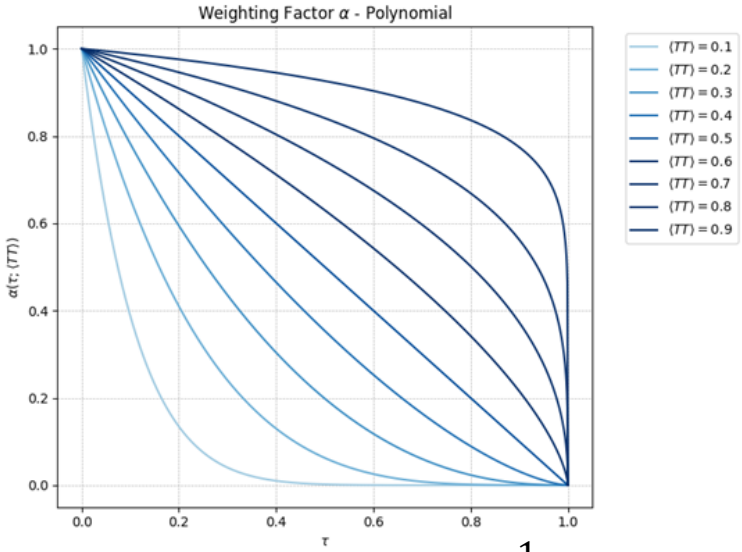
Polynomial Implementation

The parameter $\langle \mathrm{TT} \rangle$ expresses the **total weight** assigned to the **Transit Time**.

Enables a more **flexible approach** and supports more complex forms.

Recovers the **linear** formulation for $\langle \mathrm{TT} \rangle = 0.5$





$$\alpha(\tau; \langle TT \rangle) = (1-\tau)^q, \qquad q = \frac{1}{\langle TT \rangle} - 1$$

Tune $\langle \mathrm{TT} \rangle$ to **minimize** the total **estimation error** of **TFST** relative to the observed shipment times.

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Protoype & Results

Prototype Architecture

Implementation & Deployment

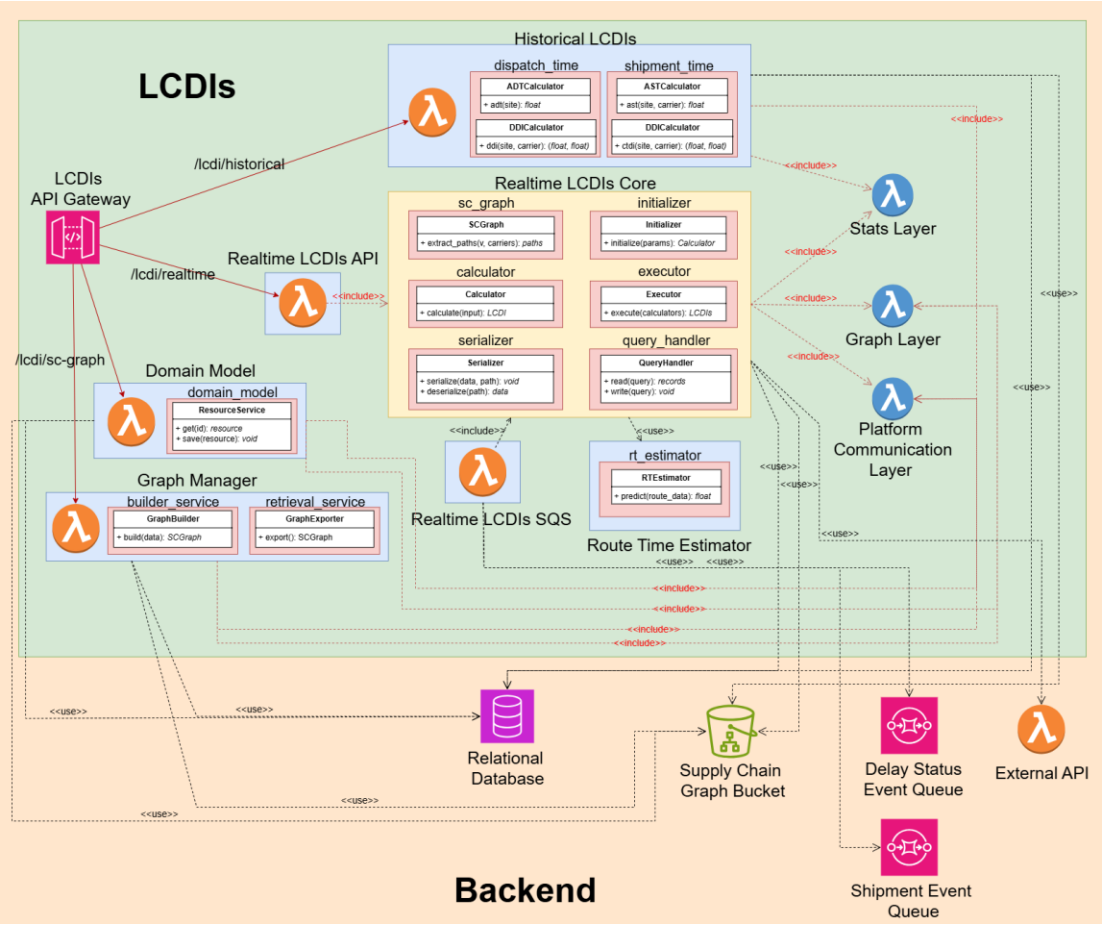
Developed in **Python** with deployment on **AWS:**

- AWS **Lambda Functions** & **Layers**.
- Amazon **RDS** (Relational Database Service).
- Amazon **S3** (Object Storage).
- Amazon **API Gateway**.
- Amazon **SQS** (Simple Queue Service).

Modularity & Configurability

Dynamic hyperparameter tuning via centralized configuration table.

Modular design enabling independent evolution of individual components.



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Evaluation Methodology (1)

Estimations Time Resolution

Distributions fitted using hourly-resolution data; model **evaluated** at **daily scale** using two approaches:

- **Hourly-based:** metrics computed from hourly predictions, then reported by day.
- **Daily-based:** predictions discretized into business days, with metrics directly evaluated at this scale.

Operational Stages

Dispatch Stage: both dispatch time and shipment time must be estimated.

Shipment Stage: dispatch time already known; only shipment time is estimated.

Set splits

Training Set: used to fit the probability distributions.

Test Set: completely unseen during training, used solely for performance evaluation.

Evaluation focused on the system's ability to accurately predict delivery time windows for orders, leveraging event data reported by carriers throughout the shipment process.

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Evaluation Methodology (2)

Evaluation steps

1. Quantitative evaluation on three metrics:
   - **Sharpness**: width of predicted intervals.
   - **Coverage:** proportion of observations within the predicted interval.
   - **Interval Score:** single measure accounting for both sharpness and coverage.
2. **Sensitivity Analysis**: assessing model coverage across varying confidence levels.
3. **Visualization** of the evolution of **shipment estimates**.

Model Variants

Multiple implementations tested for realtime indicators.

Best-performing solution:

- **Polynomial** version of the **TFST weighting factor**.
- **Baseline** approach for **DRT**.

Evaluation focused on the system's ability to accurately predict delivery time windows for orders, leveraging event data reported by carriers throughout the shipment process.

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Quantitative Evaluation

Operational Stage	Sharpness (days)		Coverage (%)		Interval Score (days)	
	Hourly	Daily	Hourly	Daily	Hourly	Daily
Dispatch	2.15	2.20	0.28	0.56	4.22	4.26
Shipment	1.03	1.18	0.52	0.84	1.45	1.63

Training Set – 90% Confidence

Operational Stage	Sharpness (days)		Coverage (%)		Interval Score (days)	
	Hourly	Daily	Hourly	Daily	Hourly	Daily
Dispatch	2.87	3.00	0.3	0.5	7.81	7.44
Shipment	1.30	1.55	0.63	0.86	1.99	2.20

Test Set – 90% Confidence

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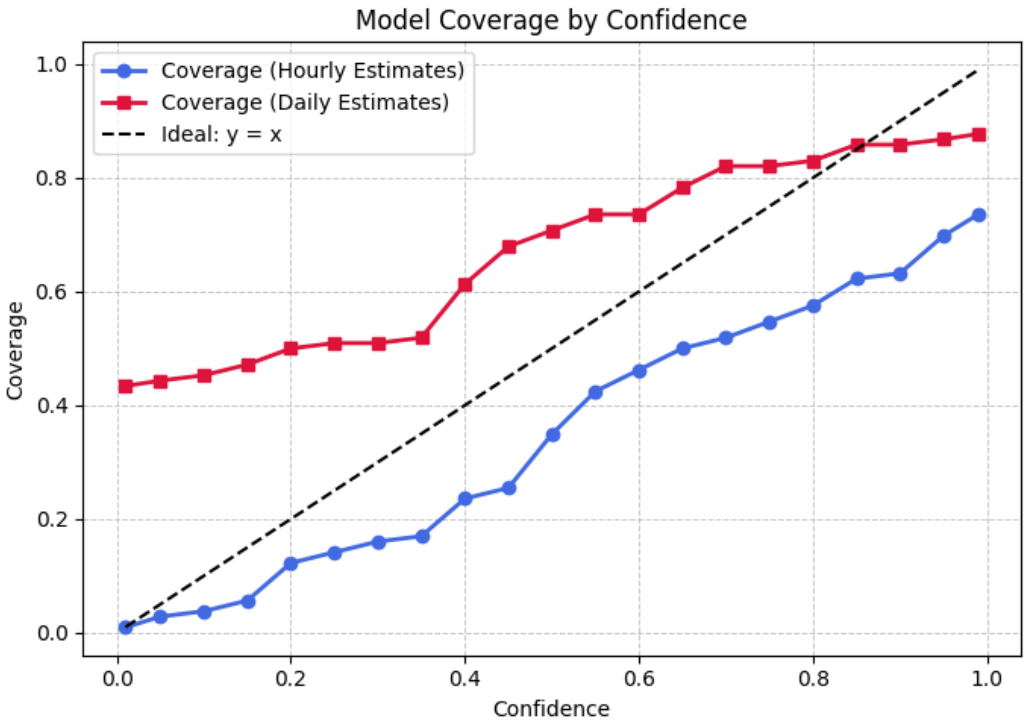
Sensitivity Analysis

Evaluation

Coverage assessed on the test set across multiple nominal confidence levels.

Hour-resolution estimates show **under-reliability**.

Day-resolution estimates show **over-reliability** up to a nominal confidence of 0.85



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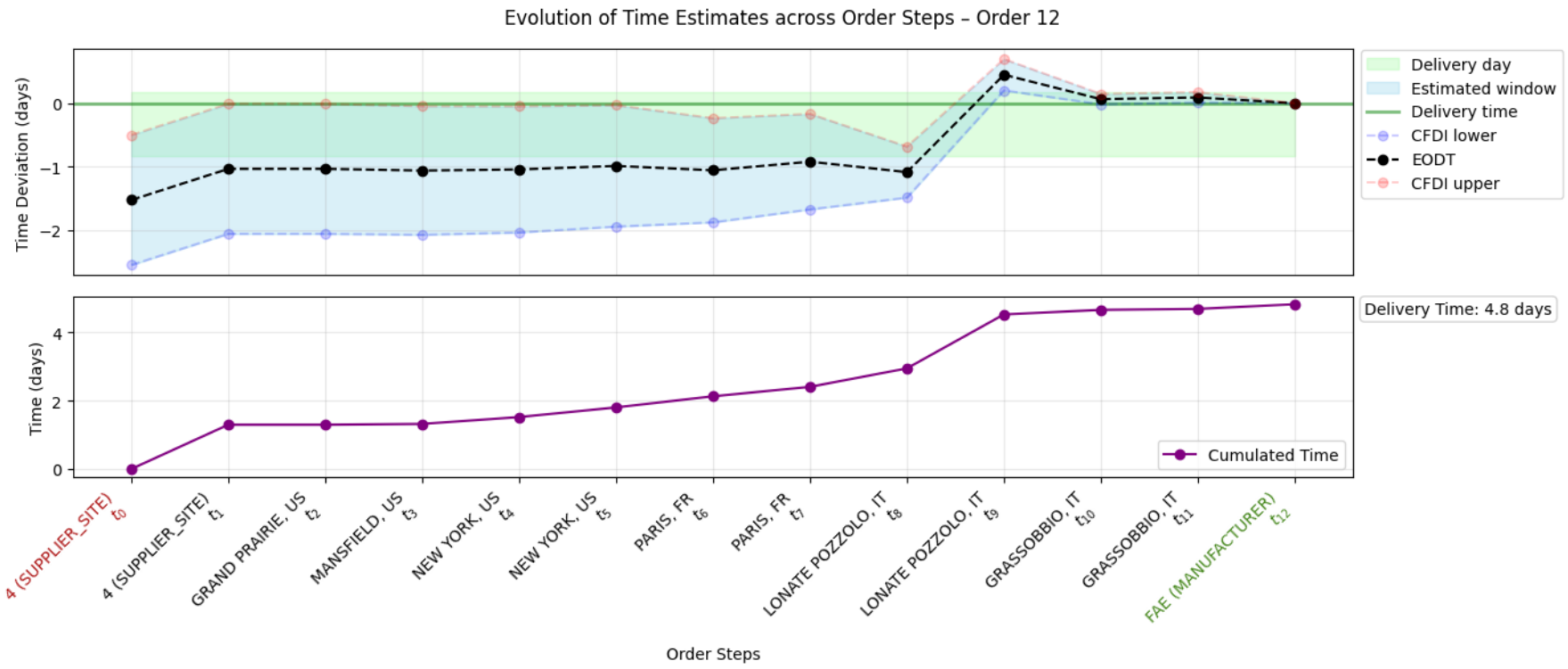
Evolution of Shipment Estimates (1)



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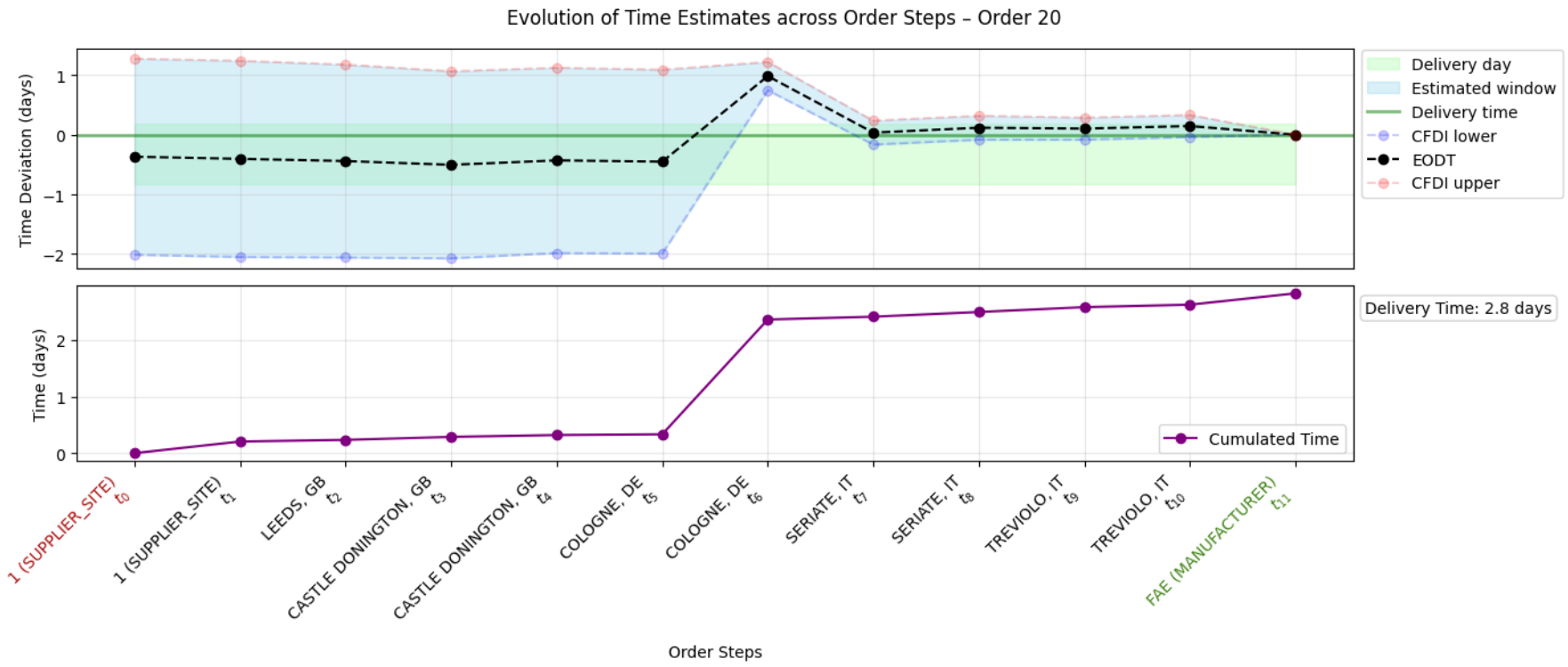
Evolution of Shipment Estimates (2)



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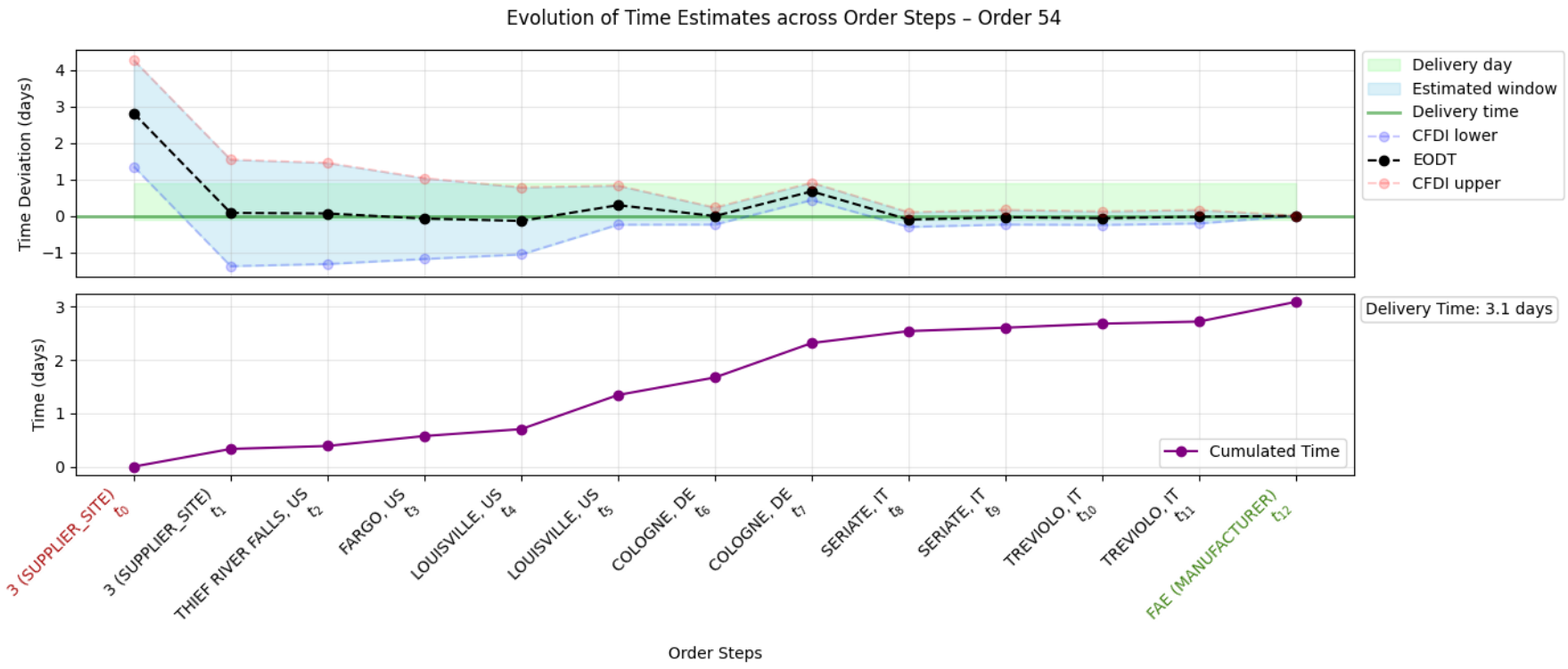
Evolution of Shipment Estimates (3)



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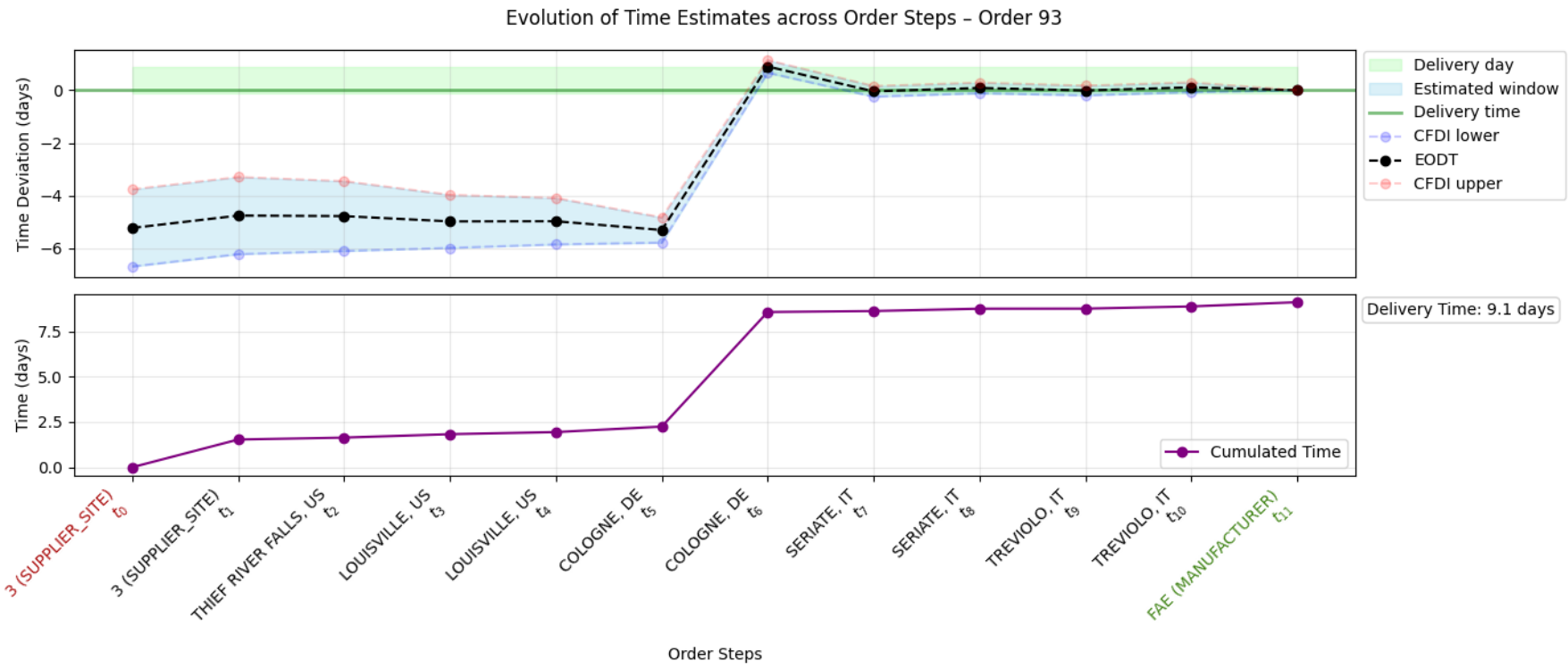
Evolution of Shipment Estimates (4)



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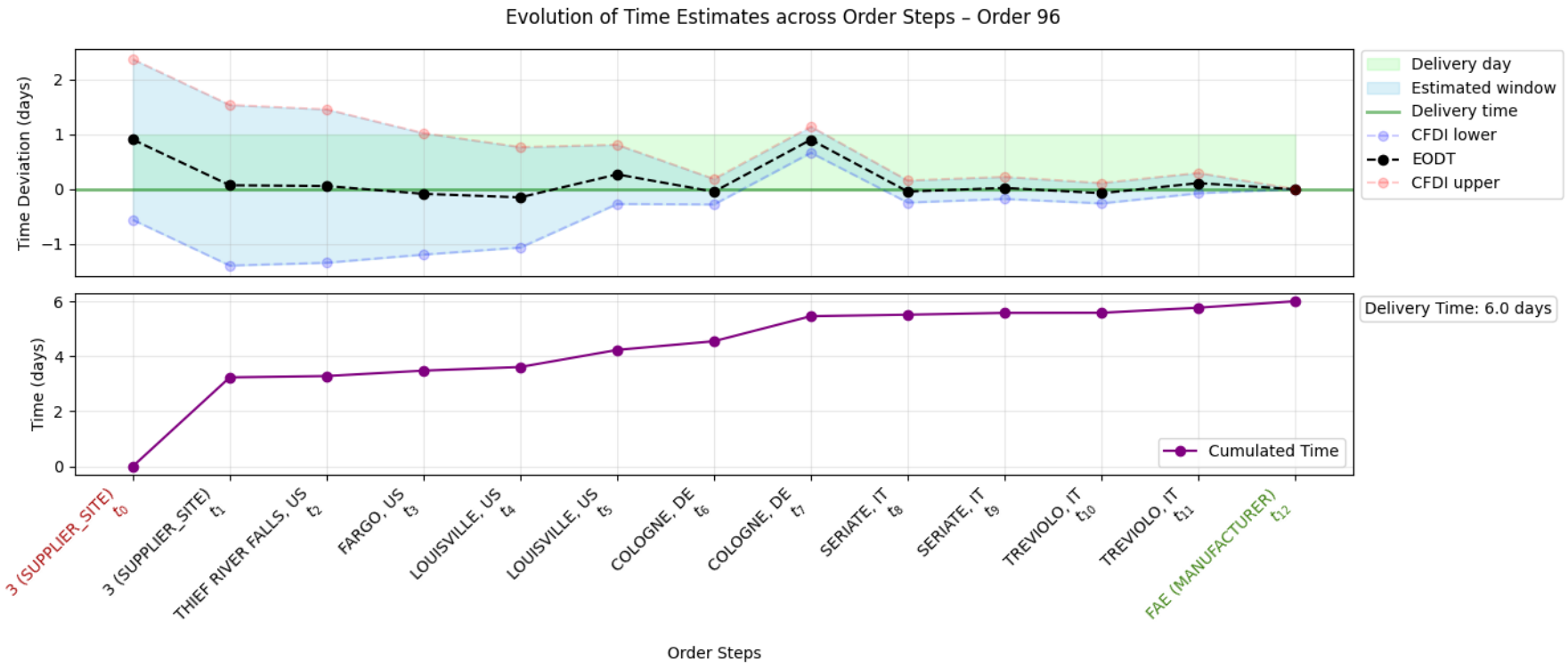
Evolution of Shipment Estimates (5)



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Conclusion

CONCLUSION

Key Findings

Real-Time Delay Detection

The system detects **disruptions** in **real time** as shipment events are received, fully meeting the project's core requirements.

Time Resolution Trade-off

Hourly estimates offer finer resolution but **lower reliability**; **daily** estimates are **better calibrated** for practical use.

Operational Stage

Dispatch stage estimates remain **unreliable**; **shipment** stage provides **actionable insights** as uncertainty decreases, even with the limited amount of data available.

Overall calibration shows some under-reliability, particularly at higher confidence levels; nevertheless, even with limited data the framework already enables effective disruption detection.

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Future Developments

Data Expansion

Expand the **dataset** and refine all data-driven components, such as probability distributions, to **improve calibration**, enhance indicators, and **increase** the **reliability** of predictions.

Refinement of Indicators Implementation

Implement and validate more advanced **solutions** for specific **indicators**, in particular **DRT** (e.g., Bayesian methods, quantile regression).

Multi-Manufacturer Extension

Extend framework to support **multiple manufacturers** and more complex operational contexts.

Future development should focus on acquiring more data to improve calibration and support the implantation of more sophisticated, formally defined solutions for specific indicators.

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